CS 1312
Algorithms Design and Analysis

Linear Programming

Introduction

- Many problems take the form of maximizing or minimizing an objective, given limited resources and competing constraints.
- If we can specify the objective as a **linear function** of certain variables, and if we can specify the constraints on resources as **equalities or inequalities** on those variables, then we have a linear programming problem.

Example

A company makes two products, A and B.

- Profit per unit:
  - Product A: $3
  - Product B: $2
- Resource limits:
  - Machine time: at most 10 hours
  - Labor time: at most 8 hours
- Resource usage per unit:
  - A uses 2 hours of machine time and 1 hour of labor
  - B uses 1 hour of machine time and 2 hours of labor

Example

- x = number of units of A
- y = number of units of B

Objective function (Maximize profit): Max Z=3x+2y

Constraints:

$2x + y \leq 10$ (Machine time)

$x + 2y \leq 8$ (Labor time)

$x \geq 0, y \geq 0$ (Non-negativity)

LP Problem:

Maximize Z=3x+2y

subject to $2x+y \leq 10$

$x+2y \leq 8$

$x,y \geq 0$

More Examples

- An oil company wants to decide where to drill for oil. Siting a drill at a particular location has an associated cost and based on geological surveys, an expected payoff of some number of barrels of oil. The company has a limited budget for locating new drills and wants to maximize the amount of oil it expects to find, given this budget.

More Examples

- An airline wishes to schedule its flight crews. The Federal Aviation Administration imposes many constraints, such as limiting the number of consecutive hours that each crew member can work and insisting that a particular crew work only on one model of aircraft during each month. The airline wants to schedule crews on all of its flights using as few crew members as possible.

Standard Form

In ***standard form***, we are given n real numbers $c_1, c_2, \ldots, c_n$; m real numbers $b_1, b_2, \ldots, b_m$; and mn real numbers a_{ij} for $i = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, n$. We wish to find n real numbers $x_1, x_2, \ldots, x_n$ that

$$\text{maximize} \quad \sum_{j=1}^{n} c_j x_j \qquad \text{(Constraints)}$$

subject to

$$\sum_{j=1}^{n} a_{ij} x_j \leq b_i \quad \text{for } i = 1, 2, \ldots, m$$

$$x_j \geq 0 \quad \text{for } j = 1, 2, \ldots, n .$$

(Constraints)

(Nonnegativity constraints)

Standard Form

In the standard form:

- the objective function is **maximization**, not minimization,
- all constraints are of the form **≤**, and
- all variables are **nonnegative**.

An overview of linear programming

- Let us first consider the following linear program with two variables:

$$\text{maximize } x_1 + x_2$$

$$\text{subject to}$$

$$4x_1 - x_2 \leq 8$$

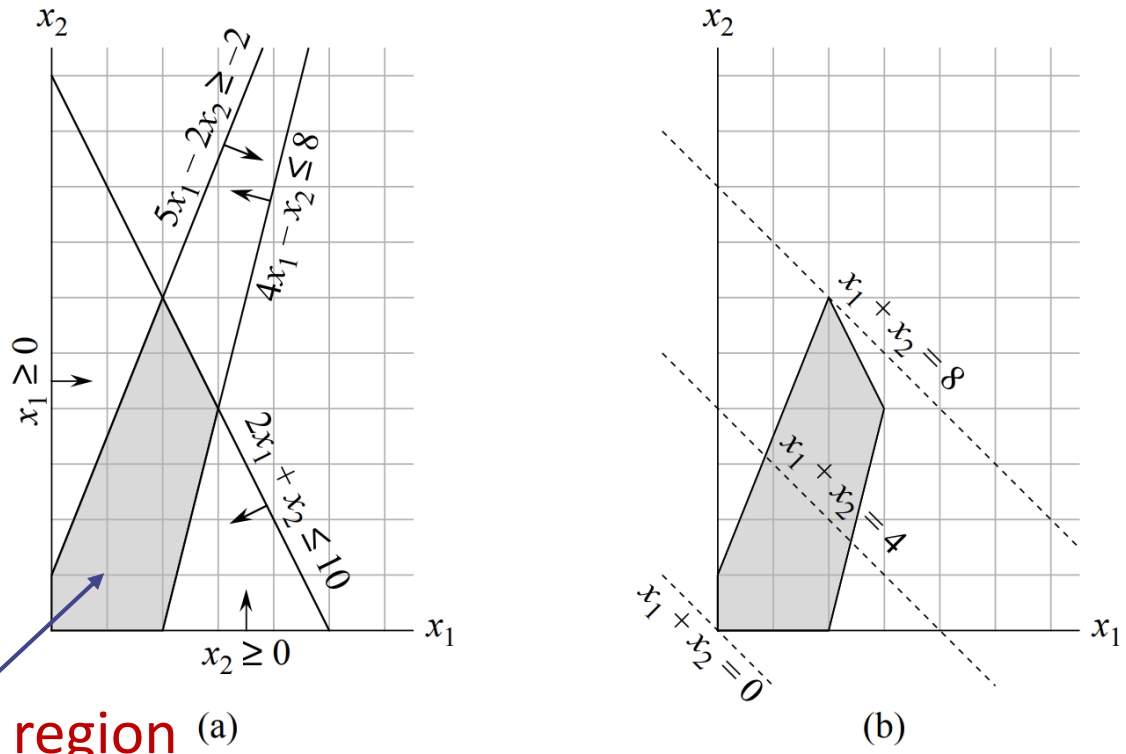
$$2x_1 + x_2 \leq 10$$

$$5x_1 - 2x_2 \geq -2$$

$$x_1, x_2 \geq 0$$

- Any setting of the variables x_1 and x_2 that satisfies all the constraints represents a feasible solution to the linear program.

An overview of linear programming



Feasible region

Any point that has the **maximum** objective value as an **optimal** solution.

Figure 29.2 **(a)** The linear program given in (29.12)–(29.15). Each constraint is represented by a line and a direction. The intersection of the constraints, which is the feasible region, is shaded. **(b)** The dotted lines show, respectively, the points for which the objective value is 0, 4, and 8. The optimal solution to the linear program is $x_1 = 2$ and $x_2 = 6$ with objective value 8.

Slack Form

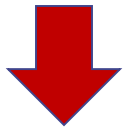
- To efficiently solve a linear program with the simplex algorithm, we prefer to express it in a form in which some of the constraints are **equality constraints**.
  - convert it into a form in which the nonnegativity constraints are the only inequality constraints, and the remaining constraints are equalities.

$$\sum_{j=1}^{n} a_{ij} x_j \le b_i \quad \Rightarrow \quad \begin{aligned} s &= b_i - \sum_{j=1}^{n} a_{ij} x_j \,, \\ s &\ge 0 \,. \end{aligned}$$

s is called a slack variable because it measures the slack or difference, between the left-hand and right-hand sides of equation.

Slack Form

$$
\begin{aligned}
\text{maximize} \quad & 2x_1 - 3x_2 + 3x_3 \\
\text{subject to} \quad & \\
& x_1 + x_2 - x_3 \le 7 \\
& -x_1 - x_2 + x_3 \le -7 \\
& x_1 - 2x_2 + 2x_3 \le 4 \\
& x_1, x_2, x_3 \ge 0 \ .
\end{aligned}
$$



$$
\begin{aligned}
\text{maximize} \quad & 2x_1 - 3x_2 + 3x_3 \\
\text{subject to} \quad & \\
& x_4 = 7 - x_1 - x_2 + x_3 \\
& x_5 = -7 + x_1 + x_2 - x_3 \\
& x_6 = 4 - x_1 + 2x_2 - 2x_3 \\
& x_1, x_2, x_3, x_4, x_5, x_6 \ge 0 \ .
\end{aligned}
$$

x_4, x_5, x_6 are called slack variables

Simplex Algorithm

- The simplex algorithm is the classical method for solving linear programs.
- The running time of the simplex algorithm is not polynomial in the worst case; however, it is often remarkably fast in practice.

Simplex Algorithm

- Basic ideas: (walking on the boundary)

  Step 1: Start at a vertex.

  Step 2: If current vertex is optimal, STOP! Otherwise,

  Step 3: Move to a better neighboring vertex. Go To Step 2.

Simplex Algorithm – Example

- A company uses three operations to assemble three types of toys: **trains**, **trucks**, and **cars**. The daily time available for these three operations is **430**, **460**, and **420** minutes, respectively.
- The unit revenues (profit per toy unit) for trains, trucks, and cars are **$3, $2**, and **$5**, respectively. The assembly times for trains in the three operations are **1, 3, and 1** minutes. The assembly times corresponding to trucks are **2, 0, and 4** minutes, and for cars they are **1, 2, and 0** minutes (a zero time indicates that the operation is not used).

Simplex Algorithm – Example

- Let X_1, X_2, and X_3 be the daily quantities of trains, trucks, and cars assembled, respectively. The information is summarized in the following table:

Operation	Daily available time in minutes			Available time per day (minutes)
	Train	Truck	Car	
1	1	2	1	430 min
2	3	0	2	460 min
3	1	4	0	420 min
Profits per unit	3	2	5	

maximize $3x_1 + 2x_2 + 5x_3$

subject to

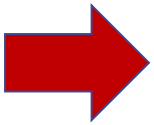
$$x_1 + 2x_2 + x_3 \leq 430$$
$$3x_1 + 0x_2 + 2x_3 \leq 460$$
$$x_1 + 4x_2 + 0x_3 \leq 420$$
$$x_1, x_2, x_3 \geq 0$$

Simplex Algorithm – Example

maximize $z = 3x_1 + 2x_2 + 5x_3$

subject to

$$x_1 + 2x_2 + x_3 \leq 430$$
$$3x_1 + 0x_2 + 2x_3 \leq 460$$
$$x_1 + 4x_2 + 0x_3 \leq 420$$
$$x_1, x_2, x_3 \geq 0$$



maximize $z - 3x_1 - 2x_2 - 5x_3 + 0s_1 + 0s_2 + 0s_3 = 0$

subject to

$$x_1 + 2x_2 + x_3 + s_1 + 0s_2 + 0s_3 = 430$$
$$3x_1 + 0x_2 + 2x_3 + 0s_1 + s_2 + 0s_3 = 460$$
$$x_1 + 4x_2 + 0x_3 + 0s_1 + 0s_2 + s_3 = 420$$
$$x_1, x_2, x_3, s_1, s_2, s_3 \geq 0$$

Simplex Algorithm – Example

maximize $z-3x_1 - 2x_2 - 5x_3+0s_1 +0s_2 +0s_3 =0$

subject to

$$x_1 +2x_2 +x_3 +s_1 +0s_2 +0s_3 = 430$$
$$3x_1 +0x_2 + 2x_3 +0s_1 +s_2 +0s_3 = 460$$
$$x_1 + 4x_2 + 0x_3 +0s_1 +0s_2 +s_3 = 420$$
$$x_1, x_2 , x_3, s_1, s_2 , s_3 \geq 0$$

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b
z	1	-3	-2	-5	0	0	0	0
s_1	0	1	2	1	1	0	0	430
s_2	0	3	0	2	0	1	0	460
s_3	0	1	4	0	0	0	1	420

Simplex Algorithm – Example

1. Select the pivot column
   - The column with the most negative value in row 1: (-5)

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b
z	1	-3	-2	-5	0	0	0	0
s_1	0	1	2	1	1	0	0	430
s_2	0	3	0	2	0	1	0	460
s_3	0	1	4	0	0	0	1	420

Simplex Algorithm – Example

2. Select the pivot row.
   - The row with the **smallest positive** ratio (b/ pivot column): 230

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b	*ratio*
z	1	-3	-2	-5	0	0	0	0	0/-5 =0
s_1	0	1	2	1	1	0	0	430	430/1=430
s_2	0	3	0	2	0	1	0	460	460/2=230
s_3	0	1	4	0	0	0	1	420	420/0

Simplex Algorithm – Example

2. Select the pivot row.
   - The row with the **smallest positive** ratio (b/ pivot column)

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b	*ratio*
z	1	-3	-2	-5	0	0	0	0	0/-5 =0
s_1	0	1	2	1	1	0	0	430	430/1=430
s_2	0	3	0	2	0	1	0	460	460/2=230
s_3	0	1	4	0	0	0	1	420	420/0

Pivot element

Simplex Algorithm – Example

3. Convert the pivot element to one.
   - Multiply the pivot row by ½.

entering variable

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b
z	1	-3	-2	-5	0	0	0	0
s_1	0	1	2	1	1	0	0	430
s_2	0	3	0	2	0	1	0	460
s_3	0	1	4	0	0	0	1	420

Leaving variable

Pivot element

Simplex Algorithm – Example

3. Convert the pivot element to one.
   - Multiply the pivot row by ½.

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b
z	1	-3	-2	-5	0	0	0	0
s_1	0	1	2	1	1	0	0	430
x_3	0	3/2	0	1	0	½	0	230
s_3	0	1	4	0	0	0	1	420

Simplex Algorithm – Example

4. Convert the other element in the pivot column to zeros.

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b	
z	1	-3	-2	-5	0	0	0	0	R_1
s_1	0	1	2	1	1	0	0	430	R_2
x_3	0	3/2	0	1	0	½	0	230	R_3
s_3	0	1	4	0	0	0	1	420	R_4

Simplex Algorithm – Example

4. Convert the other element in the pivot column to zeros.

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b	
z	1	9/2	-2	0	0	5/2	0	1150	R_1
s_1	0	1	2	1	1	0	0	430	R_2
x_3	0	3/2	0	1	0	½	0	230	R_3
s_3	0	1	4	0	0	0	1	420	R_4

$R_1 = R_1 + 5R_3$

Simplex Algorithm – Example

4. Convert the other element in the pivot column to zeros.

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b	
z	1	9/2	-2	0	0	5/2	0	1150	R_1
s_1	0	-½	2	0	1	-½	0	200	R_2
x_3	0	3/2	0	1	0	½	0	230	R_3
s_3	0	1	4	0	0	0	1	420	R_4

$R_1 = R_1 + 5R_3$

$R_2 = R_2 - R_3$

Simplex Algorithm – Example

4. Convert the other element in the pivot column to zeros.

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b	
z	1	9/2	-2	0	0	5/2	0	1150	R_1
s_1	0	-½	2	0	1	-½	0	200	R_2
x_3	0	3/2	0	1	0	½	0	230	R_3
s_3	0	1	4	0	0	0	1	420	R_4

$R_1 = R_1 + 5R_3$

$R_2 = R_2 - R_3$

$R_4 = R_4$

Simplex Algorithm – Example

1. Select the pivot column
   - The column with the most negative value in row 1: (-2)

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b
z	1	9/2	-2	0	0	5/2	0	1150
s_1	0	-½	2	0	1	-½	0	200
x_3	0	3/2	0	1	0	½	0	230
s_3	0	1	4	0	0	0	1	420

Simplex Algorithm – Example

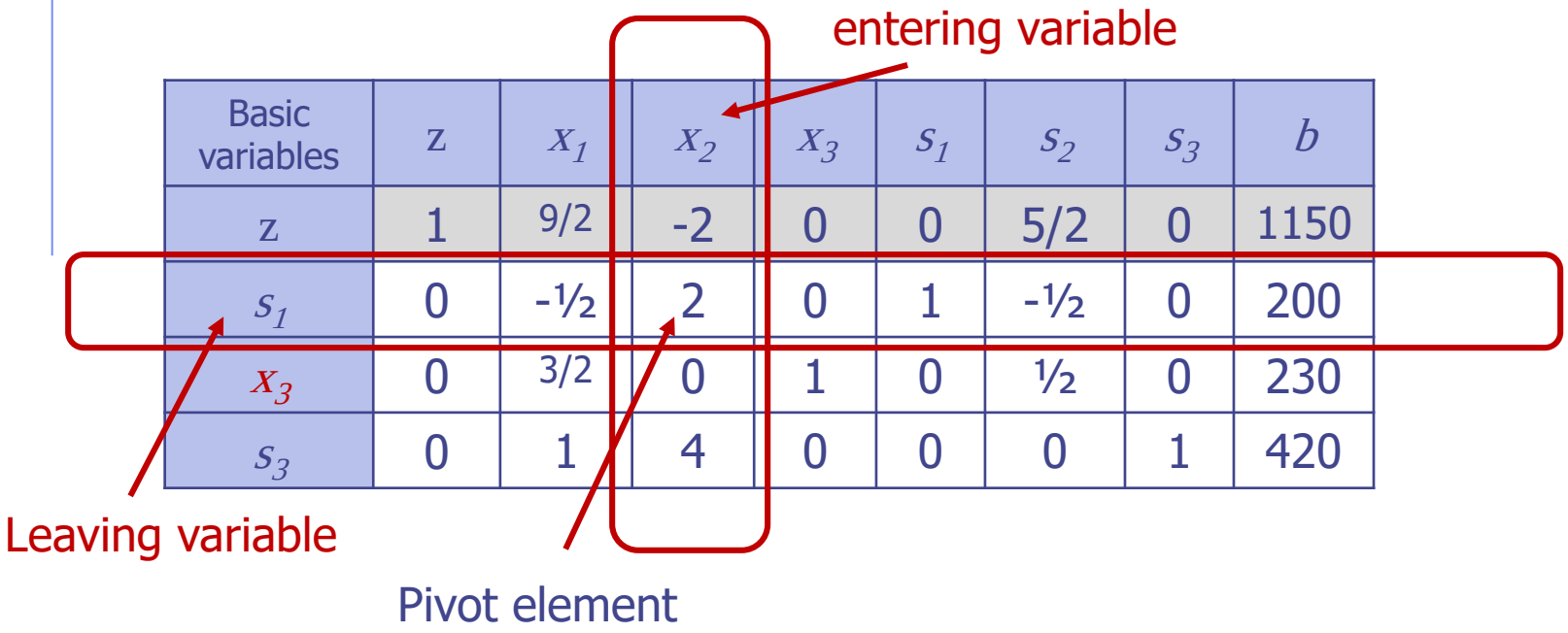
2. Select the pivot row.
   - The row with the **smallest positive** ratio (b/ pivot column)

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b	*ratio*
z	1	9/2	-2	0	0	5/2	0	1150	1150/-2
s_1	0	-½	2	0	1	-½	0	200	200/2=100
x_3	0	3/2	0	1	0	½	0	230	230/0
s_3	0	1	4	0	0	0	1	420	320/4=105

Pivot element

Simplex Algorithm – Example

3. Convert the pivot element to one.
   - Multiply the pivot row by ½.



Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b
z	1	9/2	-2	0	0	5/2	0	1150
s_1	0	-½	2	0	1	-½	0	200
x_3	0	3/2	0	1	0	½	0	230
s_3	0	1	4	0	0	0	1	420

Simplex Algorithm – Example

3. Convert the pivot element to one.
   - Multiply the pivot row by ½.

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b
z	1	9/2	-2	0	0	5/2	0	1150
x_2	0	-1/4	1	0	½	-1/4	0	100
x_3	0	3/2	0	1	0	½	0	230
s_3	0	1	4	0	0	0	1	420

Simplex Algorithm – Example

4. Convert the other element in the pivot column to zeros.

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b	
z	1	9/2	-2	0	0	5/2	0	1150	R_1
x_2	0	-1/4	1	0	½	-1/4	0	100	R_2
x_3	0	3/2	0	1	0	½	0	230	R_3
s_3	0	1	4	0	0	0	1	420	R_4

Simplex Algorithm – Example

4. Convert the other element in the pivot column to zeros.

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b	
z	1	4	0	0	1	2	0	1350	R_1
x_2	0	-1/4	1	0	½	-1/4	0	100	R_2
x_3	0	3/2	0	1	0	½	0	230	R_3
s_3	0	1	4	0	0	0	1	420	R_4

$R_1 = R_1 + 2R_2$

Simplex Algorithm – Example

4. Convert the other element in the pivot column to zeros.

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b	
z	1	4	0	0	1	2	0	1350	R_1
x_2	0	-1/4	1	0	½	-1/4	0	100	R_2
x_3	0	3/2	0	1	0	½	0	230	R_3
s_3	0	1	4	0	0	0	1	420	R_4

$R_1 = R_1 + 2R_2$

$R_3 = R_3$

Simplex Algorithm – Example

4. Convert the other element in the pivot column to zeros.

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b	
z	1	4	0	0	1	2	0	1350	R_1
x_2	0	-1/4	1	0	½	-1/4	0	100	R_2
x_3	0	3/2	0	1	0	½	0	230	R_3
s_3	0	2	0	0	-2	1	1	20	R_4

$R_1=R_1+2R_2$

$R_3=R_3$

$R_4=R_4-4R_2$

Simplex Algorithm – Example

Stop, since no negative number in **R1**.

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b
z	1	4	0	0	1	2	0	1350
x_2	0	-1/4	1	0	½	-1/4	0	100
x_3	0	3/2	0	1	0	½	0	230
s_3	0	2	0	0	-2	1	1	20

R_1

Simplex Algorithm – Example

Stop, since no negative number in **R1**.

x_1 =0 (trains), x_2 =100 (buses), x_3 =230 (cars)

Basic variables	z	x_1	x_2	x_3	s_1	s_2	s_3	b
z	1	4	0	0	1	2	0	1350
x_2	0	-1/4	1	0	½	-1/4	0	100
x_3	0	3/2	0	1	0	½	0	230
s_3	0	2	0	0	-2	1	1	20

Maximum proft z (objective value) =$ 1350 ($2x_2 + 5x_3 = 1350$)

Simplex Algorithm – Example 2

maximize z= $x_1 + 4x_2$

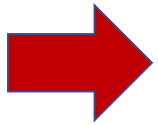
subject to

$2x_1 + x_2 \leq 3$

$3x_1 + 5x_2 \leq 9$

$x_1 + 3x_2 \leq 5$

$x_1, x_2 \geq 0$



maximize z- $x_1 - 4x_2 + 0s_1 + 0s_2 + 0s_3 = 0$

subject to

$2x_1 + x_2 + s_1 + 0s_2 + 0s_3 = 3$

$3x_1 + 5x_2 + 0s_1 + s_2 + 0s_3 = 9$

$x_1 + 3x_2 + 0s_1 + 0s_2 + s_3 = 5$

$x_1, x_2, s_1, s_2, s_3 \geq 0$

Simplex Algorithm – Example 2

maximize $z - x_1 - 4x_2 + 0s_1 + 0s_2 + 0s_3 = 0$

subject to

$$2x_1 + x_2 + s_1 + 0s_2 + 0s_3 = 3$$

$$3x_1 + 5x_2 + 0s_1 + s_2 + 0s_3 = 9$$

$$x_1 + 3x_2 + 0s_1 + 0s_2 + s_3 = 5$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0$$

Basic variables	z	x_1	x_2	s_1	s_2	s_3	b
z	1	-1	-4	0	0	0	0
s_1	0	2	1	1	0	0	3
s_2	0	3	5	0	1	0	9
s_3	0	1	3	0	0	1	5

Simplex Algorithm – Example 2

1. Select the pivot column
   - The column with the most negative value in row 1: (-4)

Basic variables	z	x_1	x_2	s_1	s_2	s_3	b
z	1	-1	-4	0	0	0	0
s_1	0	2	1	1	0	0	3
s_2	0	3	5	0	1	0	9
s_3	0	1	3	0	0	1	5

Simplex Algorithm – Example 2

2. Select the pivot row.
   - The row with the smallest positive ratio (b/ pivot column): 1.67

Basic variables	z	x_1	x_2	s_1	s_2	s_3	b	Ratio
z	1	-1	-4	0	0	0	0	0/-4=0
s_1	0	2	1	1	0	0	3	3/1=3
s_2	0	3	5	0	1	0	9	9/5=1.8
s_3	0	1	3	0	0	1	5	5/3=1.67

Simplex Algorithm – Example 2

3. Convert the pivot element to one.
   - Multiply the pivot row by 1/3.

entering variable

Basic variables	z	x_1	x_2	s_1	s_2	s_3	b
z	1	-1	-4	0	0	0	0
s_1	0	2	1	1	0	0	3
s_2	0	3	5	0	1	0	9
s_3	0	1	3	0	0	1	5

Leaving variable

Pivot element

Simplex Algorithm – Example 2

3. Convert the pivot element to one.
   - Multiply the pivot row by 1/3.

Basic variables	z	x_1	x_2	s_1	s_2	s_3	b
z	1	-1	-4	0	0	0	0
s_1	0	2	1	1	0	0	3
s_2	0	3	5	0	1	0	9
x_2	0	1/3	1	0	0	1/3	5/3

Simplex Algorithm – Example 2

4. Convert the other element in the pivot column to zeros.

Basic variables	z	x_1	x_2	s_1	s_2	s_3	b	
z	1	-1	-4	0	0	0	0	R_1
s_1	0	2	1	1	0	0	3	R_2
s_2	0	3	5	0	1	0	9	R_3
x_2	0	1/3	1	0	0	1/3	5/3	R_4

Simplex Algorithm – Example 2

4. Convert the other element in the pivot column to zeros.

Basic variables	z	x_1	x_2	s_1	s_2	s_3	b	
z	1	1/3	0	0	0	4/3	20/3	R_1
s_1	0	2	1	1	0	0	3	R_2
s_2	0	3	5	0	1	0	9	R_3
x_2	0	1/3	1	0	0	1/3	5/3	R_4

$$R_1 = R_1 + 4R_4$$

Simplex Algorithm – Example 2

4. Convert the other element in the pivot column to zeros.

Basic variables	z	x_1	x_2	s_1	s_2	s_3	b	
z	1	1/3	0	0	0	4/3	20/3	R_1
s_1	0	5/3	0	1	0	-1/3	4/3	R_2
s_2	0	3	5	0	1	0	9	R_3
x_2	0	1/3	1	0	0	1/3	5/3	R_4

$R_1 = R_1 + 4R_4$

$R_2 = R_2 - R_4$

Simplex Algorithm – Example 2

4. Convert the other element in the pivot column to zeros.

Basic variables	z	x_1	x_2	s_1	s_2	s_3	b	
z	1	1/3	0	0	0	4/3	20/3	R_1
s_1	0	5/3	0	1	0	-1/3	4/3	R_2
s_2	0	4/3	0	0	1	-5/3	2/3	R_3
x_2	0	1/3	1	0	0	1/3	5/3	R_4

$$R_1 = R_1 + 4R_4$$
$$R_2 = R_2 - R_4$$
$$R_3 = R_3 - 5R_4$$

Simplex Algorithm – Example 2

Stop, since no negative number in **R1**.

$x_1 = 0$, $x_2 = 5/3$

Basic variables	z	x_1	x_2	s_1	s_2	s_3	b
z	1	1/3	0	0	0	4/3	20/3
s_1	0	5/3	0	1	0	-1/3	4/3
s_2	0	4/3	0	0	1	-5/3	2/3
x_2	0	1/3	1	0	0	1/3	5/3

z (objective value) $= x_1 + 4x_2 = 20/3$

References

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